

Course Outcomes (COs)	Course Outcome Statements
CO1	Apply fundamental electrical laws to analyse DC circuits.
CO2	Analyse the performance of AC circuits.
CO3	Interpret the applications of basic electrical machines.
CO4	Analyse the characteristics of semiconductor devices such as diodes and transistors for its use in rectifiers, voltage regulators and amplifiers.
CO5	Explain and apply the fundamentals of digital and analog electronics, electrical measurement, sensors and transducers, power systems, and electrical safety practices including protective devices and earthing.

Mapped CO(s) Number	Relevant Major Theory Session Outcomes (TSOs)	Relevant Units/ Body of Knowledge (BoK)
CO1	<i>TSO 1a.</i> Explain classifications of electrical elements and their characteristics. <i>TSO 1b.</i> Apply Ohm's law and Kirchhoff's laws in circuit analysis. <i>TSO 1c.</i> Analyse circuits using mesh and nodal methods. <i>TSO 1d.</i> Apply divider rules and star-delta transformation in circuits.	Unit- 1.0: Basic Electrical Circuits (8hrs) 1.1 Electrical Circuit Elements: Resistance, inductance, and capacitance as active and passive elements 1.2 Classification of Elements: Unilateral and bilateral elements; linear and nonlinear elements 1.3 Basic Circuit Laws: Ohm's law and Kirchhoff's current and voltage laws 1.4 Circuit Configurations: Series and parallel circuits, voltage sources, current sources, and source transformation 1.5 Circuit Analysis Methods: Mesh analysis and nodal analysis techniques 1.6 Network Theorems and Transformations: Current divider rule, voltage divider rule, and star-delta transformation
CO2	<i>TSO 2.a</i> Explain common signals, waveforms, and AC quantities. <i>TSO 2.b</i> Apply phasor representation to analyse phase relationships in AC circuits. <i>TSO 2.c</i> Analyse impedance, admittance, power factor, and power triangle in AC circuits.	Unit- 2.0: A.C. Circuits (8hrs) 2.1 Signals and Waveforms: Common electrical signals and their waveform characteristics 2.2 AC Quantities: Instantaneous, peak, average, and RMS values of alternating signals

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	<i>TSO 2.d</i> Explain the concepts and advantages of three-phase balanced and unbalanced systems.	2.3 Waveform Factors: Peak factor and form factor of sinusoidal waves 2.4 Phase and Phasors: Phase difference, lagging, leading, in-phase quantities, and phasor representation in rectangular and polar forms 2.5 AC Circuit Parameters: Impedance, admittance, power factor, and power triangle 2.6 Three-Phase Systems: Necessity, advantages, phase sequence, and balanced/unbalanced supply and load concepts
CO3	<i>TSO 3a.</i> Explain electromagnetic laws, self-inductance, and mutual inductance. <i>TSO 3b.</i> Describe the construction, operation, and applications of DC machines. <i>TSO 3c.</i> Apply transformer equations to determine ratios and regulation. <i>TSO 3d.</i> Explain the working principles and applications of induction and fractional horsepower motors.	Unit- 3.0: Basic Electrical Machines (8hrs) 3.1 Electromagnetic Principles: Basic laws of electromagnetism, self-inductance, and mutual inductance 3.2 DC Machines: Principle of operation, construction, classification, and applications of DC machines 3.3 Single-Phase Transformers: Necessity, working principle, construction, and E.M.F. equation of transformers 3.4 Transformer Performance: Voltage ratio, current ratio, and voltage regulation with significance 3.5 Induction Motors: Overview of single-phase and three-phase induction motors and their working principles 3.6 Motor Applications: Applications of induction motors and fractional horsepower motors
CO4	<i>TSO 4a.</i> Explain intrinsic, extrinsic semiconductors, and diode characteristics. <i>TSO 4b.</i> Analyse forward/reverse bias conditions and diode breakdown mechanisms. <i>TSO 4c.</i> Apply diodes in rectifiers, filters, and voltage regulation circuits. <i>TSO 4d.</i> Explain BJT operation and its use as amplifier and switch.	Unit- 4.0: Semiconductor Devices (8hrs) 4.1 Semiconductor Materials: Intrinsic and extrinsic semiconductor types 4.2 p-n Junction Diode: Construction, characteristics, forward bias, and reverse bias operation 4.3 Diode Behavior: Temperature effects and breakdown mechanisms in diodes 4.4 Diode Applications: Half-wave rectifier, full-wave rectifier, filters, and Zener diode as voltage regulator 4.5 BJT Fundamentals: PNP and NPN transistors, construction, and operating principles 4.6 BJT Applications: CE configuration as amplifier

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		and transistor as a switch
CO5	TSO 5a Explain number systems and perform binary addition. TSO 5b Apply logic gates and Boolean algebra to solve digital logic problems. TSO 5c Describe the operation of LEDs and seven-segment displays using truth tables. TSO 5d Explain the functions and applications of digital and analog ICs, including IC 7805 and IC 7912. TSO 5e Explain the features and uses of multimeter and CRO. TSO 5f Describe the basics of sensors, transducers, and power system layout. TSO 5g Apply electrical safety practices using fuse, MCB, and MCCB. TSO 5h Explain the purpose and importance of earthing systems.	Unit- 5.0: Fundamentals of Digital, Analog Electronics, Measurement and Electrical Safety (10hrs) 5.1 Binary, octal, and hexadecimal number systems; binary addition and basic number operations 5.2 Basic logic gates, universal gates, and Boolean algebra 5.3 LEDs, seven-segment display and truth table; overview of digital and analog integrated circuits 5.4 Functions and applications of IC 7805 and IC 7912; basic features and applications of multimeter and CRO 5.5 Fundamentals, types, and uses of sensors and transducers; generation, transmission, and distribution of electrical power 5.6 Safe handling practices, prevention of electrical hazards; functions and applications of fuse, MCB, and MCCB 5.7 Purpose, importance, and methods of earthing systems

Reference/Books:

S. No.	Titles	Author(s)	Publisher and Edition with ISBN
1.	Basic Electrical Engineering	D. P. Kothari, I. J. Nagrath	McGraw-Hill Education, 2019, ISBN: 978-9353165727
2.	Electronic Device & Circuit theory	Robert L. Boylestad and Louis Nashelsky	Pearson, 2013, ISBN: 978-1292025636
3.	Digital Logic and Computer Design	M. Morris Mano	Pearson India, 2017, ISBN: 978-9332586048
4.	Electrical Machinery	P. S. Bimbhra	Khanna Publishers, 2024, ISBN: 978-9389139105

S. No.	Titles	Author(s)	Publisher and Edition with ISBN
5.	A Course in Electrical and Electronic Measurements and Instrumentation	A. K. Sawhney	Dhanpat Rai & Co., 19th Edition, ISBN: 978-8177001006
6.	Electrical Safety of Low Voltage Systems	Massimo Mitolo	McGraw-Hill, 1st Edition, 2009 ISBN: 978-0071508186
7.	Fundamentals of Electric Circuits	Charles K. Alexander and Matthew N. O. Sadiku	McGraw-Hill, 7th Edition, ISBN: 978-9355320162